



PYMBLE LADIES' COLLEGE

MATHEMATICS

TRIAL HSC EXAMINATION

2005

Reading Time: 5 minutes
Working Time: 3 hours

Instructions to students:

- Write using blue or black biro.
- All questions may be attempted.
- Diagrams are not to scale.
- All necessary working should be shown in every question.
- Your name and your teacher's name may be written before you begin the assessment.
- Start each question in a **new** booklet.
- Marks may be deducted for careless or untidy work.
- Board-approved calculators may be used.
- A table of standard integrals is provided.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

Question 1 (12 marks)

Use a separate writing booklet

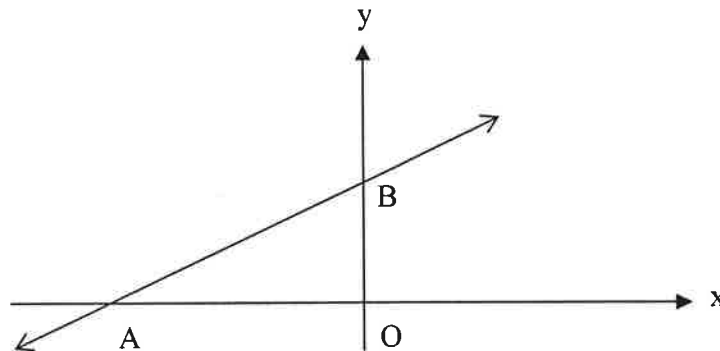
MARKS

- (a) Evaluate $\sqrt{\frac{3.705^3}{93.87 \times 2.7}}$, correct to two decimal places. **2**
- (b) Solve: $\frac{x+1}{4} - \frac{x-3}{5} = 1$ **2**
- (c) Express $\frac{\sqrt{5}-\sqrt{3}}{2-\sqrt{3}}$ with a rational denominator. **2**
- (d) If Patrick was given a discount of 25% followed by a discount of 4%, what percentage of the original price did he have to pay? **2**
- (e) Solve: $x^2 - 5x < 6$ **2**
- (f) A function is defined as $f(x) = \begin{cases} 2x-1 & \text{for } 0 \leq x \leq 3 \\ \frac{1}{3}x+4 & \text{for } 3 < x \leq 5 \end{cases}$ **2**

What is the range of this function?

Question 2 (12 marks)

Use a separate writing booklet

MARKS

A is the point $(-3, 0)$ and B is the point $(0, 2)$.

- | | | |
|--------|--|---|
| (i) | Calculate the length of the interval AB . | 1 |
| (ii) | Find the gradient of the line AB . | 1 |
| (iii) | Show that the equation of the line AB is $2x - 3y + 6 = 0$. | 1 |
| (iv) | Write down the co-ordinates of a point C such that $ABCO$ is a parallelogram. | 1 |
| (v) | Write down the co-ordinates of the point M where the diagonal AC cuts the y axis. | 1 |
| (vi) | What is the size of the acute angle (to the nearest degree) made by the line AB with the positive direction of the x axis? | 1 |
| (vii) | Hence determine the size of $\angle ABC$. | 1 |
| (viii) | Find the perpendicular distance from the origin to the line AB . | 2 |
| (ix) | Hence, or otherwise, find the area of the parallelogram $ABCO$. | 1 |
| (x) | Shade the area satisfied by the following simultaneously: | 2 |

$$x \leq 0 \text{ and } y \geq 0 \text{ and } 2x - 3y + 6 \leq 0$$

Question 3 (12 marks)

Use a separate writing booklet

MARKS(a) Differentiate with respect to x :

(i) $2 \cos 3x$

2

(ii) $5x^2(1+x)^3$

2

(b) Evaluate:

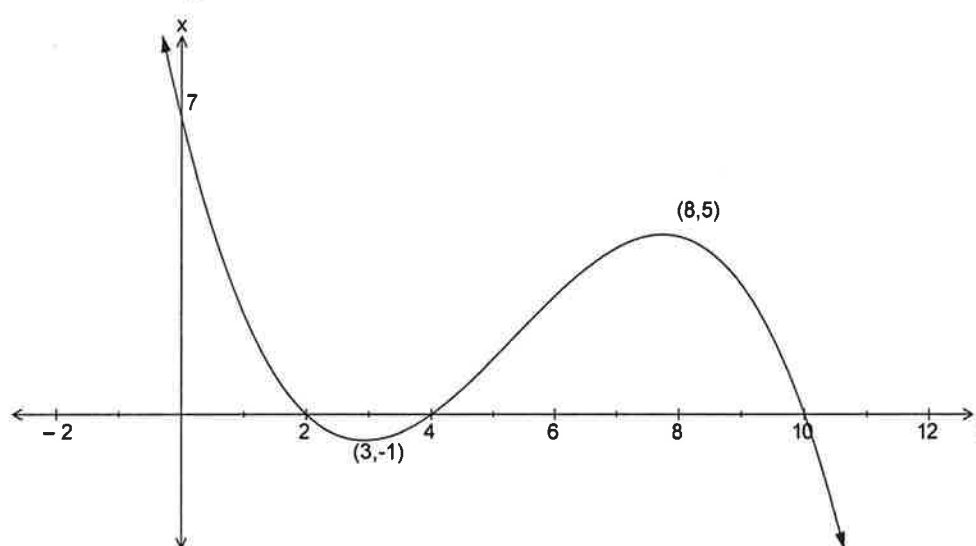
(i) $\int_1^2 \frac{2}{x} dx$

2

(ii) $\int_0^2 (e^{2x} + 1) dx$

2

(c)



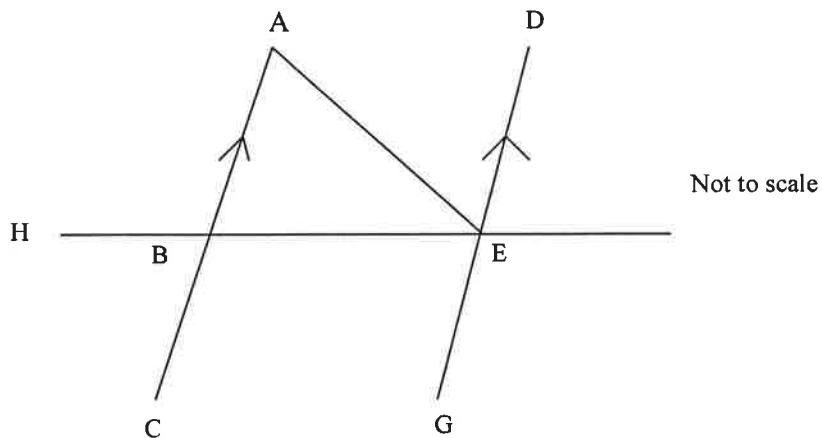
Copy the diagram of $x = f(t)$ and graph $x = f'(t)$ onto the same pair of axes.

1

Question 3 (continued)

MARKS

(d)



In the diagram, $AB = AE$, $AC \parallel DG$, $\angle ABH = 146^\circ$ and $\angle AED = x^\circ$.

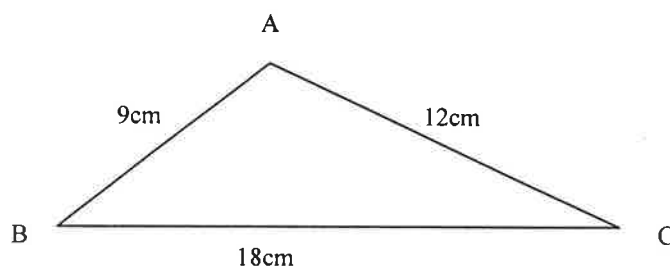
- | | | |
|------|---|----------|
| (i) | Copy this diagram into your writing booklet and place all the information onto the diagram. | 1 |
| (ii) | Find the value of x , giving complete reasons. | 2 |

Question 4 (12 marks)

Use a separate writing booklet

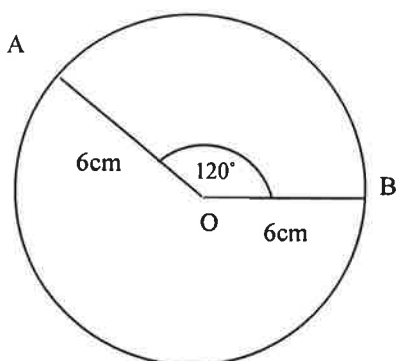
MARKS

(a)

Given the sides of a triangle ABC are 9cm, 12cm and 18cm, find:

- | | | |
|------|---|---|
| (i) | the size of the smallest angle to the nearest degree; | 2 |
| (ii) | the area of $\triangle ABC$ to the nearest square centimetre. | 2 |

(b)



Not to scale

 O is the centre of the circle of radius 6cm. $\angle AOB = 120^\circ$.

- | | | |
|------|--|---|
| (i) | Find the length of the arc AB , leaving answer in exact form. | 2 |
| (ii) | Find the area of the minor sector, leaving answer in exact form. | 2 |
- (c) If p , q and 32 are the first three terms of a geometric series and q , 4 and p are the first three terms of another geometric series, find p and q . 4

Question 5 (12 marks)

Use a separate writing booklet

MARKS

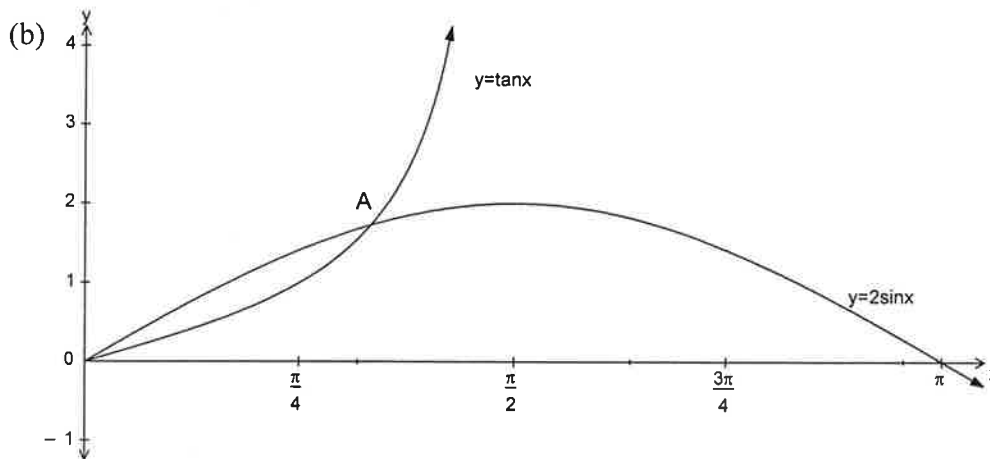
- (a) Given the curve $y = x^3 - x^2 - x + 1$ find:
- | | | |
|-------|---|----------|
| (i) | any stationary point(s) and determine their nature; | 3 |
| (ii) | any point(s) of inflexion. | 2 |
| (iii) | Sketch the curve, showing all essential features in the domain $-2 \leq x \leq 3$. | 2 |
| (iv) | Hence, find the minimum value of $x^3 - x^2 - x + 1$ for $-2 \leq x \leq 3$. | 1 |
- (b) Find the equation of the tangent to $y = \ln(3x + 1)$ at the point (2, 5). **2**
- (c) Sketch $y = 3 \sin 2x$ in the domain $0 \leq x \leq 2\pi$. **2**

Question 6 (12 marks)

Use a separate writing booklet

MARKS

- (a) Find the values of m for which the equation $2x^2 + mx + 8 = 0$ has no real roots. 2



Consider the curves $y = \tan x$ and $y = 2 \sin x$ in the domain $0 \leq x < \frac{\pi}{2}$.

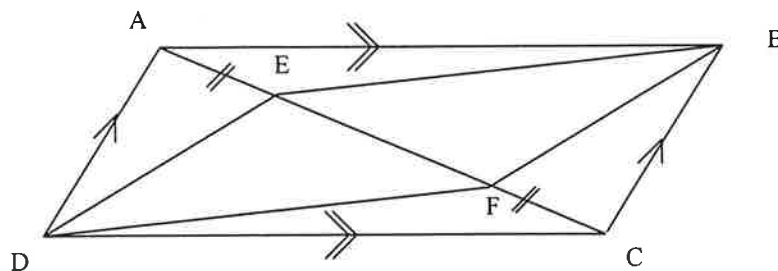
- (i) Show that the coordinates of A are $\left(\frac{\pi}{3}, \sqrt{3}\right)$. 2
- (ii) Show that $\frac{d}{dx}[\ln \cos x] = -\tan x$. 1
- (iii) Hence find the area enclosed by these two curves in the above diagram. 3
- (c) (i) Show the locus of a point which moves so that it is equidistant from the point $(0, 3)$ and the line $y = -3$ is a parabola $x^2 = 12y$. 2
- (ii) Find the vertex and the focal length of this parabola. 2

Question 7 (12 marks)

Use a separate writing booklet

MARKS

(a)



ABCD is a parallelogram.

On the diagonal AC, points E and F are chosen such that $AE = FC$.

- (i) Prove that $\triangle ADE$ is congruent to $\triangle CBF$. **2**
- (ii) Hence, or otherwise, show that the quadrilateral DEBF is a parallelogram. **2**

(b)

The position x cm at time t seconds of a particle moving in a straight line is given by $x = 5t + e^{-5t}$.

- (i) Find the position of the particle when $t = 1$. Give your answer correct to 3 significant figures. **1**
- (ii) By finding an expression for the velocity of the particle, show that initially the particle is at rest. **2**
- (iii) Find the limiting velocity of the particle as $t \rightarrow \infty$. **1**

Question 7 (continued)

MARKS

- (c) The mass M , of a radioactive substance t years after it starts to decay is given by $M = M_0 e^{-kt}$ where M is the mass in kilograms of the substance present and M_0 , k are constants.

If in three years, a mass of 12 kilograms will reduce to 8 kilograms, find:

- | | | |
|-------|---|---|
| (i) | the decaying constant k (answer to three decimal places). | 2 |
| (ii) | the number of kilograms present after ten years (answer to the nearest gram). | 1 |
| (iii) | the number of years it takes for the mass to halve itself. | 1 |

Question 8 (12 marks)

Use a separate writing booklet

MARKS

- (a) A curve $y = f(x)$ has the following properties in the interval $a \leq x \leq b$: 2
 $f(x) > 0$; $f'(x) > 0$; $f''(x) < 0$.

Sketch a curve satisfying these conditions.

- (b) Use the Trapezoidal Rule with three function values to find an 3
approximation for $\int_{0.5}^{1.5} \frac{\sin x}{x} dx$.

Give your answer correct to two decimal places.

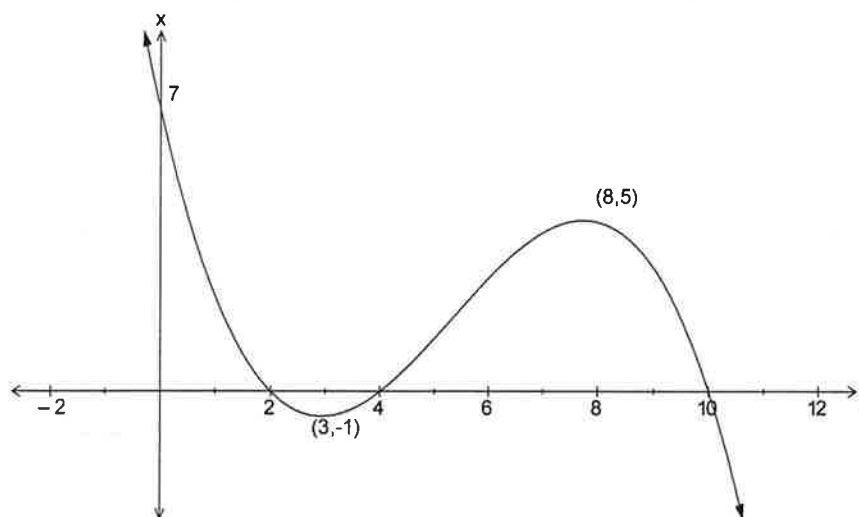
- (c) A pendulum is set swinging by lifting it to the right and releasing it. Its first swing (from right to left) is through an angle of 30° . If the next swing (from left to right) is through 27° , and each succeeding swing is through 90% of the angle of the previous swing.
- (i) List the first four angles through which the pendulum swings. 2
- (ii) Explain why these four values form a geometric sequence. 1
- (iii) Find an expression for the n^{th} term of this sequence. 1
- (iv) Which swing will be the first swing through less than 2° ? 2
- (v) Explain why it is impossible for the total angle the pendulum swings through to be greater than 300° . 1

Question 9 (12 marks)

Use a separate writing booklet

MARKS

(a)



- | | | |
|-------|--|---|
| (i) | Where was the particle initially? | 1 |
| (ii) | When was the particle at the origin? | 1 |
| (iii) | When was the particle at rest? | 2 |
| (iv) | How far did the particle travel during the first 10 seconds? | 2 |

(b) A cylindrical can without a lid is to be made from $300\pi \text{ cm}^2$ of sheet metal.

- | | | |
|------|--|---|
| (i) | Show that the volume, $V \text{ cm}^3$, may be expressed in terms of the radius, $r \text{ cm}$, by $V = 150\pi r - \frac{\pi r^3}{2}$. | 2 |
| (ii) | Hence, find the radius of the can which will give the can a maximum volume. | 4 |

Question 10 (12 marks) Use a separate writing booklet

MARKS

- (a) Stephanie wishes to have \$50 000 capital in eight years time. She invests a fixed amount of money at the beginning of each month during this time. Interest is accumulated at 6% per annum, compounded monthly.

- (i) Let $\$P$ be the monthly investment. Show that the total amount, $\$A$, after eight years is given by : **2**

$$A = \$P (1.005 + 1.005^2 + \dots + 1.005^{96})$$

- (ii) Find, to the nearest dollar, the amount to be invested each month in order to achieve her goal. **2**

- (b) Given $f(x) = \frac{e^x - e^{-x}}{2}$.

- (i) Find $f'(x)$. **1**

- (ii) Show that the graph $y = f(x)$ is increasing for all values of x and that there is a point of inflexion at $x = 0$. **3**

- (iii) Sketch the graph of $y = f(x)$. **1**

- (iv) Let $y = f(x)$. **1**
Show that this equation can be written in the form
 $e^{2x} - 2ye^x - 1 = 0$.

- (v) Hence, deduce that $x = \ln(y + \sqrt{y^2 + 1})$. **2**

END OF PAPER

Question 1

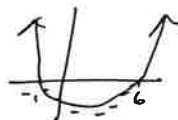
a) $\sqrt{0.20066612\dots}$
 $= 0.44795\dots$
 $= 0.45$

b) $\frac{x+1}{4} - \frac{x-3}{5} = 1$
 $20\left(\frac{x+1}{4}\right) - 20\left(\frac{x-3}{5}\right) = 20$
 $5x + 5 - 4x + 12 = 20$
 $x + 17 = 20$
 $x = 3$

c) $\frac{\sqrt{5}-\sqrt{3}}{2-\sqrt{3}} = \frac{\sqrt{5}-\sqrt{3}}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}}$
 $= \frac{(\sqrt{5}-\sqrt{3})(2+\sqrt{3})}{4-3}$
 $= 2\sqrt{5} + \sqrt{15} - 2\sqrt{3} - 3$

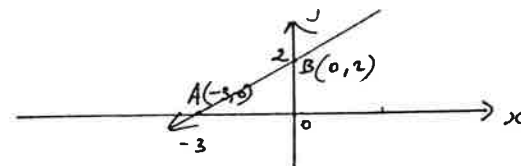
d) $\left(\frac{96}{100} \times \frac{75}{100}\right)x = 0.72x$
 $\therefore$ He pays 72% of original price of \$x.

e) $x^2 - 5x < 6$
 $x^2 - 5x - 6 < 0$
 $(x+1)(x-6) < 0$
 $\therefore -1 < x < 6$



f) When $x = 0$, $f(x) = (2 \times 0) - 1 = -1$
 When $x = 5$, $f(x) = \frac{1}{3} \times 5 + 4 = 5\frac{2}{3}$
 $\therefore$ Range: $-1 \leq f(x) \leq 5\frac{2}{3}$

Question 2



i) $AB^2 = 3^2 + 2^2$
 $= 13$
 $AB = \sqrt{13}$ units

ii) $m(AB) = \frac{2-0}{0+3} = \frac{2}{3}$

iii) Eqn of AB: $y - 2 = \frac{2}{3}(x - 0) \Rightarrow y = \frac{2}{3}x + 2$
 $3y - 6 = 2x$
 $2x - 3y + 6 = 0$

iv) $C = (3, 2)$

v) $M = (0, 1)$ midpoint of AC

vi) $m(AB) = \frac{2}{3}$
 $\therefore \tan \theta = \frac{2}{3}$
 $\therefore \theta = 34^\circ$

vii) $\angle ABC = 180^\circ - 34^\circ$

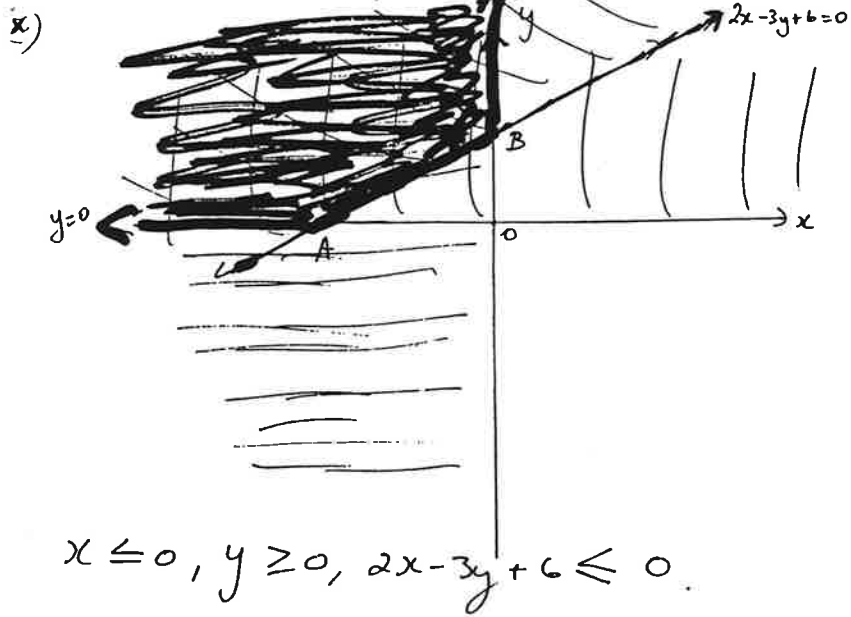
$\therefore \angle ABC = 146^\circ$

viii) $x_1 = 0, y_1 = 0, a = 2, b = -3, c = 6$

$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$
 $= \frac{|0 + 0 + 6|}{\sqrt{4 + 9}}$
 $= \frac{6}{\sqrt{13}}$ units

ix) $\text{Area}(\triangle ABC) = \frac{\sqrt{13} \times \frac{6}{\sqrt{13}}}{2}$
 $= 3$ units²

Question 2 (Contd.)



Question 3

(i) $y = 2 \cos 3x$

$$\frac{dy}{dx} = 2 \times -\sin 3x \times 3$$

$$= -6 \sin 3x$$

(ii) $y = 5x^2(1+x)^3$

$$\frac{dy}{dx} = 10x(1+x)^3 + 5x^2 \times 3(1+x)^2$$

$$= 10x(1+x)^3 + 15x^2(1+x)^2$$

$$= 5x(1+x)^2 [2(1+x) + 3x]$$

$$= 5x(1+x)^2 (2 + 2x + 3x)$$

$$= 5x(1+x)^2 (2 + 5x)$$

(b) (i) $\int_1^2 \frac{2}{x} dx = 2 [\ln x]_1^2$

$$= 2(\ln 2 - \ln 1)$$

$$= 2 \ln 2$$

(ii) $\int_0^2 (e^{2x} + 1) dx$

$$= \left[\frac{e^{2x}}{2} + x \right]_0^2$$

$$= \left(\frac{e^4}{2} + 2 \right) - \left(\frac{e^0}{2} + 0 \right)$$

$$= \frac{e^4}{2} + 2 - \frac{1}{2}$$

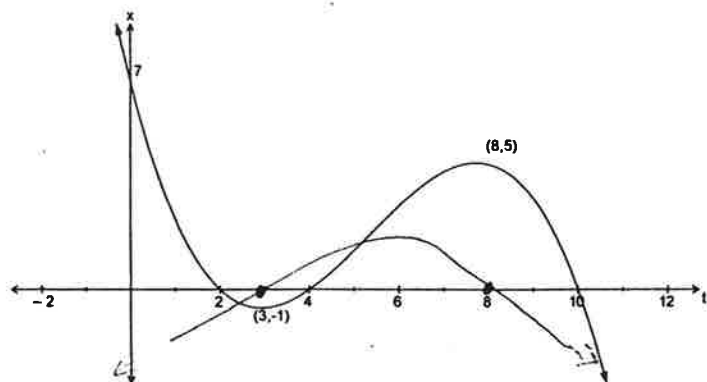
$$= \frac{e^4}{2} + 1\frac{1}{2}$$

or

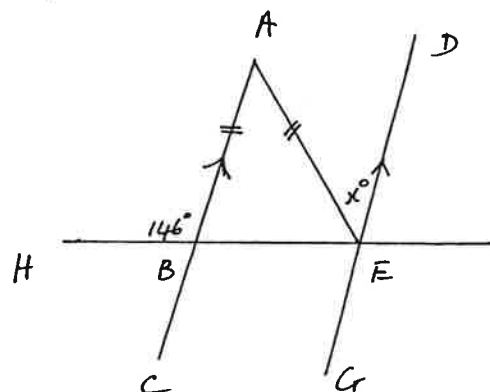
$$\frac{e^4 + 3}{2}$$

Question 3 (contd.)

(c)



(d)



$$\angle ABE = 34^\circ \text{ (supp. adj. to } \angle ABH)$$

$$\angle AEB = \angle ABE \text{ (base angles of isos. triangle)}$$

$$\therefore \angle ABE = 34^\circ$$

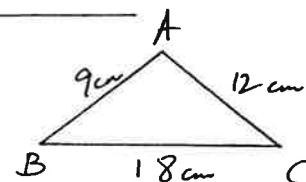
$$\angle BED = \angle ABH \text{ (corr. angles, } AC \parallel DG)$$

$$\therefore x + 34 = 146$$

$$\therefore x = 146 - 34 = 112$$

Question 4

(a)



(i) smallest angle opposite smallest side.

$$\cos C = \frac{18^2 + 12^2 - 9^2}{2 \times 18 \times 12}$$

$$= 0.8958333 \dots$$

$$\therefore C = 26^\circ$$

(ii) $\text{Area}(\triangle ABC) = \frac{1}{2} \times 18 \times 12 \times \sin C$

$$= 108 \sin 26^\circ$$

$$= 47 \text{ cm}^2$$

(b) (i) $\text{arc AB} = \frac{120}{360} \times 2\pi \times 6$

$$= 4\pi \text{ cm}$$

or

$$l = r\theta$$

$$= 6 \times \frac{2\pi}{3}$$

$$= 4\pi \text{ cm.}$$

(ii) $\text{Area} = \frac{1}{2} r^2 \theta$

$$= \frac{1}{2} \times 6^2 \times \frac{2\pi}{3}$$

$$= 12\pi \text{ cm}^2$$

or $\frac{\theta}{360} \times \pi r^2$

$$= \frac{120}{360} \times \pi \times 6^2$$

Question 4 (Contd)

(c) $p, q, 32$... GP -- (1)
 $q, 4, p$... GP -- (2)

(1) $\frac{p}{q} = \frac{32}{q}$
 $q^2 = 32p$... (1)

(2) $\frac{4}{q} = \frac{p}{4}$
 $pq = 16$... (2)

From (2) $p = \frac{16}{q}$

Subst in (1)

$$q^2 = 32 \times \frac{16}{q}$$

$$q^3 = 512$$

$$q = 8$$

$$\therefore p = \frac{16}{8} = 2$$

Question 5

(a) $y = x^3 - x^2 - x + 1$

$$y' = 3x^2 - 2x - 1$$

$$y'' = 6x - 2$$

(i) Stat pts when $y' = 0$

$$3x^2 - 2x - 1 = 0$$

$$(3x+1)(x-1) = 0$$

$$\therefore x = -\frac{1}{3}, 1$$

When $x = -\frac{1}{3}$
 $y = 1\frac{5}{27}$
 When $x = 1$
 $y = 0$

When $x = -\frac{1}{3}$
 $y'' = 6x - 2 = -4 < 0$ $\therefore$ a max tp. at $(-\frac{1}{3}, 1\frac{5}{27})$

When $x = 1$
 $y'' = 6 - 2 = 4 > 0$ $\therefore$ a min tp. at $(1, 0)$

(b) Pt. of inflexion

$$6x - 2 = 0$$

$$6x = 2$$

$$x = \frac{1}{3}$$

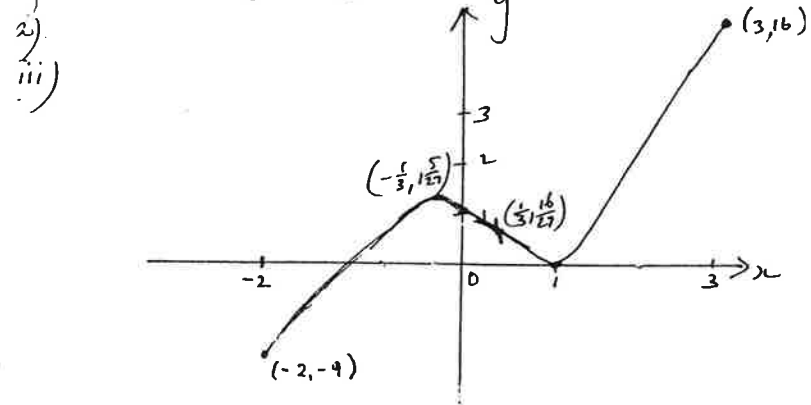
When $x = \frac{1}{3}$
 $y = \frac{14}{27}$

x	$\frac{1}{3}^-$	$\frac{1}{3}$	$\frac{1}{3}^+$
y''	-	0	+

$\therefore$ a change in concavity

$\therefore$ a point of inflexion at $(\frac{1}{3}, \frac{14}{27})$

Question 5 (contd.)



When $x = -2$, $y = (-2)^3 - (-2)^2 + 2 + 1$
 $= -9$

When $x = 3$, $y = (3)^3 - (3)^2 - 3 + 1$
 $= 16$

iv). $\therefore$ Minimum is -9 when $x = -2$
 (value)

7) $y = \ln(3x+1)$
 $y' = \frac{3}{3x+1}$

$m(\text{tangent at } (2, 5)) = \frac{3}{6+1} = \frac{3}{7}$

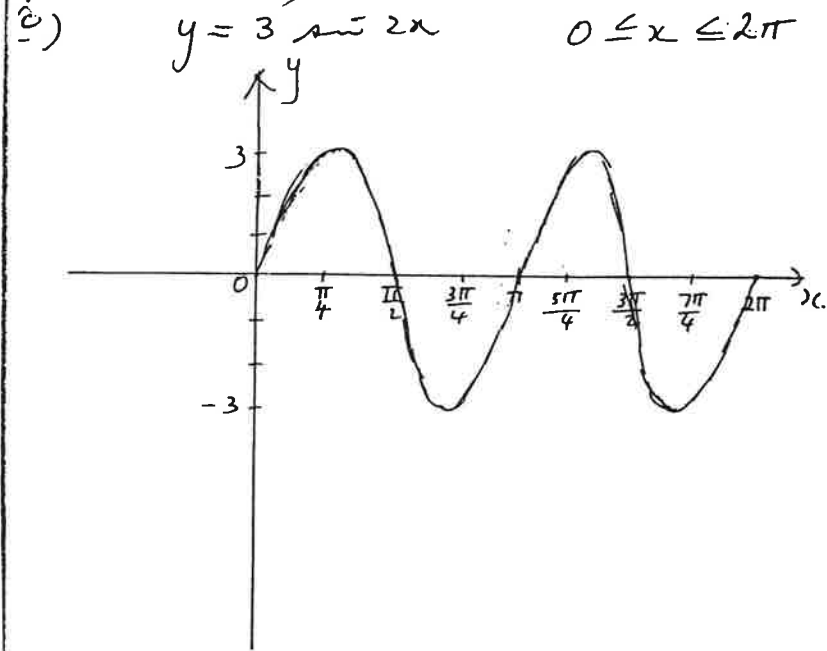
$\therefore$ Eqn. of tangent with $m = \frac{3}{7}$, thro. $(2, 5)$

is: $y - 5 = \frac{3}{7}(x - 2)$

$7y - 35 = 3x - 6$

$3x - 7y + 29 = 0$

Question 5 (contd.)



Question b

(a) $2x^2 + mx + 8 = 0$

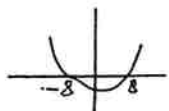
[2] No real roots $\Rightarrow b^2 - 4ac < 0$

$$b^2 - 4ac = m^2 - 4 \times 8 \times 2$$

$$= m^2 - 64$$

For $m^2 - 64 < 0$ or $m^2 < 64$

$$-8 < m < 8 //$$



(b) $\begin{cases} y = \tan x - 0, & 0 \leq x < \frac{\pi}{2} \\ y = 2 \sin x - 0 \end{cases}$

[2] (i) Sub $(\frac{\pi}{3}, \sqrt{3})$ and $(0,0)$ into both the equations or solve simultaneously

①, $\sqrt{3} = \tan \frac{\pi}{3}$
 $\sqrt{3} = \sqrt{3}$ (True)

② $\sqrt{3} = 2 \sin \frac{\pi}{3}$
 $\sqrt{3} = 2 \left(\frac{\sqrt{3}}{2} \right)$
 $\sqrt{3} = \sqrt{3}$ (True)

$$\tan x = 2 \sin x$$

$$\frac{\sin x}{\cos x} - 2 \sin x = 0$$

$$\sin x \left(\frac{1}{\cos x} - 2 \right) = 0$$

$$\sin x = 0 \text{ or } \cos x = \frac{1}{2}$$

$$x = 0 \text{ or } x = \frac{\pi}{3}$$

$$\therefore y = 0 \text{ or } y = \tan \frac{\pi}{3} = \sqrt{3}$$

Hence $(\frac{\pi}{3}, \sqrt{3})$ is a pt of inter on the 2 graphs

(ii) Show $\frac{d}{dx} [\ln \cos x] = -\tan x$

[1] $\frac{d}{dx} [\ln \cos x] = \frac{1}{\cos x} (-\sin x)$
 $= \frac{-\sin x}{\cos x} = -\tan x$

[3] (ii) Area =

$$= \int_0^{\frac{\pi}{3}} (2 \sin x - \tan x) dx$$

$$= [-2 \cos x + \ln \cos x]_0^{\frac{\pi}{3}}$$

$$= (-2 \cos \frac{\pi}{3} + \ln \cos \frac{\pi}{3}) - (-2 \cos 0 + \ln \cos 0)$$

$$= (-2 \times \frac{1}{2} + \ln \frac{1}{2}) - (-2 + \ln 1)$$

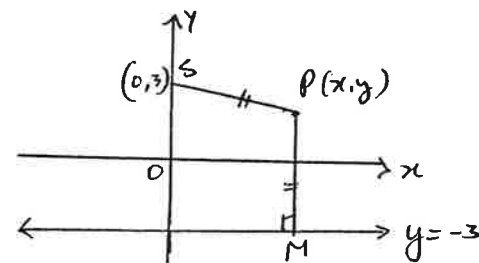
$$= -1 + \ln \frac{1}{2} + 2$$

$$= (1 + \ln \frac{1}{2}) \text{ s.u.}$$

or = $1 - \ln 2$

(c) (i)

[2]



Given $PM = PS$

ie. $y + 3 = \sqrt{x^2 + (y-3)^2}$

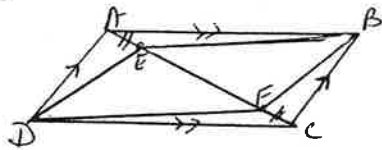
$$y^2 + 6y + 9 = x^2 + y^2 - 6y + 9$$

Hence $x^2 = 12y$ — locus of P.

[2] (ii) Vertex is $(0,0)$
 focal length = 3 units.

Question 1

(a)



2

- (i) In $\triangle ADE$ and $\triangle CBF$
 $AE = FC$ (given)
 $AD = BC$ (Opp sides of parall. ABCD)
 $\angle DAE = \angle BCF$ (Alt. angles, $AD \parallel BC$)
 $\therefore \triangle ADE \cong \triangle CBF$ (SAS)

- 2 (ii) Let $\angle AED = \angle BFC = x$ (Corr. $\angle$ s of cong. $\triangle$ s ADE and CBF)

$$\therefore \angle DEF = 180^\circ - x \text{ (Straight } \angle \text{)}$$

$$\angle BFE = 180^\circ - x \text{ (")}$$

$$\text{Hence } \angle DEF = \angle BFE$$

$$\therefore ED \parallel BF \text{ (Since alt. } \angle \text{s DEF \& BFE are equal.)}$$

$$\text{Also, } ED = BF \text{ (Corr. sides of cong. } \triangle \text{s ADE \& CBF)}$$

$$\therefore DEBF \text{ is a parallelogram. (1 pair of opposite equal and } \parallel \text{).}$$

(b) $x = 5t + e^{-5t}$

4

- (i) When $t=1$,

$$x = 5 + e^{-5} = 5.0067 \\ = 5.01 \text{ (3 sig fig.)}$$

(ii) $V = \frac{dx}{dt} = 5 - 5e^{-5t}$

$$\text{When } t=0, V = 5 - 5e^0 \\ = 5 - 5 \\ = 0$$

Hence initially the particle is at rest.

- (iii) As $t \rightarrow \infty$, $V = 5 - \frac{5}{e^{5t}}$ as e^{5t} becomes very large.
 $\rightarrow 5$
 $\therefore \text{Limiting vel} = 5 \text{ cm/s}$

$$M = M_0 e^{-kt}$$

- (i) When $t=3$, $M_0=12$, $M=8$

2 $8 = 12 e^{-3k}$

$$e^{-3k} = \frac{8}{12} = \frac{2}{3}$$

$$-3k = \ln \frac{2}{3}$$

$$\therefore k = \frac{\ln \frac{2}{3}}{-3} = 0.135 \text{ (3 d.p.)}$$

- (ii) When $t=10$, find M .

1 $M = 12 e^{-0.135 \times 10}$
 $= 3.106 \text{ kg OR } 3106 \text{ g}$

- (iii) When $M=6$, find t .

1 $6 = 12 e^{-0.135t}$
 $e^{-0.135t} = \frac{1}{2}$

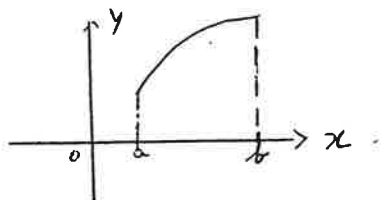
$$-0.135t = \ln \frac{1}{2}$$

$$t = \frac{\ln \frac{1}{2}}{-0.135} = 5 \text{ yrs. (to nearest whole no.)}$$

Question 2

(a)

[2]



(b)

[3]

$$\int_{0.5}^{1.5} \frac{\sin x}{x} dx$$

x	0.5	1	1.5
$f(x)$	0.9589	0.8415	0.6650

$$\int_{0.5}^{1.5} \frac{\sin x}{x} dx \div \frac{0.5}{2} [0.9589 + 2 \times 0.8415 + 0.6650]$$

$$= \underline{\underline{0.83}} \text{ (2 d.p.)}$$

(c)

[7]

(i) $30^\circ, 27^\circ, 24.3^\circ, 21.87^\circ$

(ii) It has a common ratio of 90% (or 0.9).

$$(iii) T_n = ar^{n-1}$$

$$= 30 \times 0.9^{n-1}$$

(iv) Let $T_n < 2$, find n .

$$30 \times 0.9^{n-1} < 2$$

$$0.9^{n-1} < \frac{1}{15}$$

$$n-1 > \frac{\ln \frac{1}{15}}{\ln 0.9} \div 25.70$$

$$\therefore n > 25.70 + 1$$

$$n > 26.70$$

OR use
Trial and Error

The 27th swing //

$$(v) S_\infty = \frac{a}{1-r} = \frac{30^\circ}{1-0.9} = 300^\circ$$

This implies the total angle the pendulum swings thro is always less than 300°

Question 1

(a) (i) 7 m to the right of the origin.

[6]

(ii) At 2, 4 and 10 seconds.

(iii) At $t=3$ sec and 8 sec.

$$(iv) 8 + 6 + 5 = 19 \text{ m.}$$

$$\begin{array}{c} \uparrow \quad \uparrow \quad \uparrow \\ \left(\frac{1}{2}\right) \quad (1) \quad \left(\frac{1}{2}\right) \end{array}$$

OR

$$7 + 1 + 1 + 5 + 5 = 19 \text{ m}$$

$$\begin{array}{c} \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \\ \left(\frac{1}{2}\right) \quad \left(\frac{1}{2}\right) \quad \left(\frac{1}{2}\right) \quad \left(\frac{1}{2}\right) \end{array}$$

9. (b) (i) Given surface area = 300π .

[2] $A = \pi r^2 + 2\pi r h$

$\therefore 300\pi = \pi(r^2 + 2rh)$

$h = \frac{300 - r^2}{2r}$

$V = \pi r^2 h$

$V = \pi r \cdot \frac{300 - r^2}{2}$
 $= 150\pi r - \frac{\pi r^3}{2} \text{ cm}^3$

(ii) $\frac{dV}{dr} = 150\pi - \frac{3\pi r^2}{2}$

[4] For S.P., let $\frac{dV}{dr} = 0$.

ie. $150\pi - \frac{3\pi r^2}{2} = 0$.

$\frac{3\pi r^2}{2} = 150\pi$

$r^2 = \frac{300}{2} = 150$

ie. $r = 10$ since $r > 0$.

$\frac{d^2V}{dr^2} = -\frac{3\pi}{2} \cdot 2r = -3\pi r$

When $r = 10$, $\frac{d^2V}{dr^2} = -30\pi < 0$.

$\therefore$ Max V occurs when $r = 10$ cm.

Question 10

(a) (i) Interest = $\frac{6\%}{12}$ per month = 0.005 / month.

[2] Let A_n be the amount accumulated after n months

$A_1 = P(1.005)$

$A_2 = P(1.005)^2 + P(1.005) = P(1.005^2 + 1.005)$

$A_3 = P(1.005^3 + 1.005^2 + 1.005)$

$\vdots$

$A_{96} = P(1.005^{96} + 1.005^{95} + \dots + 1.005)$

If A is the total amount accumulated, then

$A_{96} = A = P(1.005 + 1.005^2 + \dots + 1.005^{96}) //$

(ii) Let $A = \$50000$

[2] ie. $50000 = P(1.005 + 1.005^2 + \dots + 1.005^{96})$
 G.P. with
 $a = 1.005, r = 1.005, n = 96$

$\therefore 50000 = P \cdot \frac{1.005[1 + 1.005^{96}]}{1.005 - 1}, S_n = \frac{a(1-r^n)}{1-r}$

$50000 = P \cdot \frac{0.6172\dots}{0.005} \div 123.44 P$

$\therefore P = \$405 //$ to the nearest dollar.

Alternative solution

a) Let A_n = the final amount of the n^{th} investment.

$\therefore A_1 = P(1.005)^{96}$

$A_2 = P(1.005)^{95}$

$\vdots$

$A_{96} = P(1.005)^1$

$\therefore A = A_1 + A_2 + \dots + A_{96}$

$= P(1.005)^{96} + \dots + P(1.005)^1$

$= P(1.005^{96} + \dots + 1.005^1)$

10 (b) $f(x) = \frac{e^x - e^{-x}}{2}$

1 (1) $f'(x) = \frac{e^x + e^{-x}}{2}$

3 (ii) Since $e^x > 0$ for all x
 $e^{-x} > 0$ for all x

$\therefore f'(x) = \frac{e^x + e^{-x}}{2} > 0$ for all x

$\therefore y = f(x)$ is an increasing fn.

$f''(x) = \frac{e^x - e^{-x}}{2}$

For pt of inflexion, let $f''(x) = 0$.

ie. $\frac{e^x - e^{-x}}{2} = 0$

$e^x - e^{-x} = 0$

$e^x = e^{-x}$

$x = -x$

$2x = 0$

$x = 0, y = f(0) = \frac{e^0 - e^0}{2} = 0$

OR
 $f''(0) = \frac{e^0 - e^0}{2} = 0$

Checking:

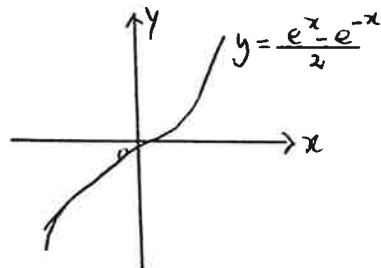
x	0^-	0	0^+
$f'(x)$	$-$	0	$+$

Since there is a change of concavity

$\therefore (0,0)$ is a pt of inflexion.

(iii)

1



(iv) Let $y = \frac{e^x - e^{-x}}{2}$

1

$2y = e^x - e^{-x}$

Multiply by e^x ,

$2ye^x = e^{2x} - e^0$

$2ye^x = e^{2x} - 1$

$\therefore e^{2x} - 2ye^x - 1 = 0$

OR $2y = e^x - \frac{1}{e^x}$

$2y = \frac{e^{2x} - 1}{e^x}$

(v) Solving $(e^x)^2 - 2y(e^x) - 1 = 0$

2 using quadratic formula,
 $a=1, b=-2y, c=-1$

$e^x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{2y \pm \sqrt{4y^2 + 4}}{2}$

$= \frac{2y \pm 2\sqrt{y^2 + 1}}{2} = y \pm \sqrt{y^2 + 1}$

But $e^x > 0$ for all x

$\therefore e^x = y + \sqrt{y^2 + 1}$ only

Taking log to base e on both sides,

$\ln e^x = \ln [y + \sqrt{y^2 + 1}]$

$x \ln e = \ln [y + \sqrt{y^2 + 1}]$

$x = \ln [y + \sqrt{y^2 + 1}]$